Course Name: Database Management System

Course Outcome

- CO1: Understand the concept of database management systems and Relational database.
- CO2: Identify the various data model used in database design.
- CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
- CO4: Create and populate a RDBMS for a real life application, with constraints and keys using SQL.
- CO5: Select the information from a database by formulating complex queries in SQL.
- CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
- CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
- CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 04 **University Roll No. .....**

End Term Examination, Odd Semester, 2023-24
B. Tech. (CSE), Year: II, Semester: III
BCSC1003: DATABASE MANAGEMENT SYSTEM

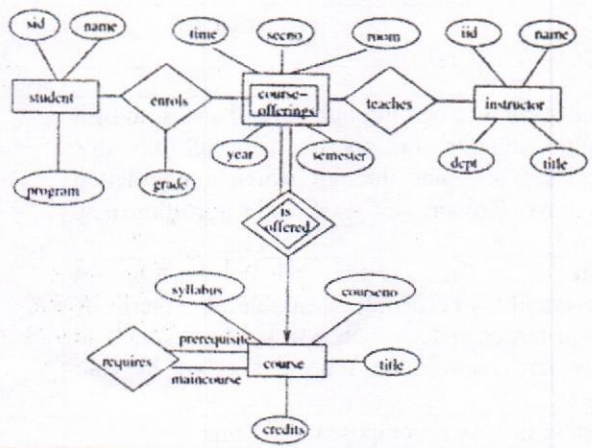
Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. Attempt all questions
2. Don't switch between the section A, B and Section C, first solve one section then move to second
3. Use only blue pen to answer
4. No B sheet will be given

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the 3-Level architecture of Database Management Systems using proper diagram.	4	CO1	U	C
2	Construct appropriate tables for the given E-R diagram. 	4	CO2	C	P

3	Given a relational Schema R (A, B, C, D) and set of Function Dependency FD = {B→A, AD→BC, C→ABD}. Find the canonical cover?	4	CO3	E	P
4	Consider the following two transactions: T1: read (*A*); read (*B*); **if** A = 0**then** *B*: = *B* + 1; write(*B*). T2: read (*B*); read (*A*); **if** B = 0**then** *A*: = *A* + 1; write (*A*). Add lock and unlock instructions to transactions T1 and T2, so that they observe the two-phase locking protocol.	4	CO5	A	P
5	Sometimes a transaction has to be undone after it has committed, because it was erroneously executed, for example because of erroneous input by a bank teller. Give an example to show that using the normal transaction undo mechanism to undo such a transaction could lead to an inconsistent state.	4	CO5	An	F

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	A transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur.	3	CO5	U	F
7	What is a recoverable schedule? Why is recoverability of schedules desirable? Are there any circumstances under which it would be desirable to allow non recoverable schedules? Explain your answer.	3	CO8	An	C
8	Suppose that we decompose the schema R = (A, B, C, D, E) into (A, B, C) and (A, D, E).	3	CO6	E	M

	Show that this decomposition is a lossless-join decomposition if the following set F of functional dependencies holds: {A → BC, CD → E, B → D, E → A}				
9	Define Boyce-Codd normal form. How does it differ from 3NF? Why is it considered a stronger form of 3NF?	3	CO6	U	C
10	Check whether the given schedule is conflict serializable or not? If not, is it view equivalent? **S** **T1** | **T2** | **T3** R (A) W (A) W (A) W (A)	3	CO7	An	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Consider the concurrent execution of two transactions, T1 & T2 as: **T1** | **T2** R (x) x = x – N R (x) x = x + M W (x) R (y) W (x) y = y + N W (y)	5	CO7	E	P

	The initial values of x, y, M & N are 200, 100, 10 & 20 respectively. Now determine: (a). What would be the values of x & y at the end of the execution of T1 & T2? (b). If the transactions are executed in serial manner; what would be the values of x & y at the end of the serial execution of T1 & T2?				
12	What is a fragment of a relation? What are the main types of fragments? Why is fragmentation a useful concept in distributed database design?	5	CO8	U	C
13	Consider the following relations for an order-processing application database at ABC, Inc. ORDER (O#, Odate, Cust#, Total_amount) ORDER_ITEM(O#, I#, Qty_ordered, Total_price, Discount%) Assume that each item has a different discount. The Total_price refers to one item, Odate is the date on which the order was placed, and the Total_amount is the amount of the order. If we apply a natural join on the relations ORDER_ITEM and ORDER in this database, what does the resulting relation schema look like? What will be its key? Show the FDs in this resulting relation. Is it in 2NF? Is it in 3NF? Why or why not? (State assumptions, if you make any.)	5	CO6	C	P